Exercise 1. Suppose both $\sum x_n$ and $\sum y_n$ converge absolutely. Show that the Cauchy product, $\sum z_n$ where $z_n = x_0 y_n + x_1 y_{n-1} + \cdots + x_n y_0$, also converges absolutely.

Proof. Since $\sum |x_n|$ and $\sum |y_n|$ are convergent in $\mathbb{R}$, $\sum |x_n|$ and $\sum |y_n|$ are bounded in $\mathbb{R}$. Then $\exists X, Y \in \mathbb{R}$ such that

$$\forall m \in \mathbb{N} : \sum_{n=0}^{m} |x_n| \leq X$$

and

$$\forall m \in \mathbb{N} : \sum_{n=0}^{m} |y_n| \leq Y.$$

By the Triangle Inequality for $\mathbb{R}$,

$$\begin{aligned}
\forall m \in \mathbb{N} : \sum_{n=0}^{m} |z_n| &= \sum_{n=0}^{m} \left| \sum_{k=0}^{n} x_k y_{n-k} \right| \\
&\leq \sum_{n=0}^{m} \sum_{k=0}^{n} |x_k y_{n-k}| \\
&= |x_0 y_0| + (|x_0 y_1| + |x_1 y_0|) + \cdots + (|x_0 y_m| + |x_1 y_{m-1}| + \cdots + |x_m y_0|) \\
&= \sum_{n=0}^{m} |x_n| \cdot \sum_{k=0}^{m-n} |y_k| \leq XY.
\end{aligned}$$

Thus, $\sum |z_n|$ is bounded and increasing (because $|z_n| \geq 0 \ \forall n \in \mathbb{N}$) in $\mathbb{R}$; hence, $\sum |z_n|$ converges in $\mathbb{R}$. □

Exercise 2. Find all real numbers x so that the power series converges.

(a)
$$\sum_{n=0}^{\infty} 2^n x^n$$

(b)
$$\sum_{n=0}^{\infty} n x^n$$

(c)
$$\sum_{n=0}^{\infty} \frac{1}{(2n)!} (x-10)^n$$

(d)
$$\sum_{n=0}^{\infty} n! x^n$$

Proof. Start!

(a) By the Root Test,

$$x \in \left(-\frac{1}{2}, \frac{1}{2} \right) \iff \lim_{n\to\infty} \sqrt[n]{|2^n x^n|} \left(= \lim_{n\to\infty} \sqrt[n]{|2x|^n} = 2 \lim_{n\to\infty} |x| = 2|x| \right) < 1$$

and

$$\sum_{n=0}^{\infty} |2^n x^n| < \infty \implies \sum_{n=1}^{\infty} 2^n x^n < \infty.$$

By the Geometric Series Test,

$$x = -\frac{1}{2} \implies \sum_{n=0}^{\infty} 2^n x^n = \sum_{n=1}^{\infty} (-1)^n \not< \infty$$

and

$$x = \frac{1}{2} \implies \sum_{n=0}^{\infty} 2^n x^n = \sum_{n=0}^{\infty} 2^n = \infty.$$

(b) By the Root Test,

$$x \in (-1,1) \iff \lim_{n\to\infty} \sqrt[n]{|nx^n|} \left(= \lim_{n\to\infty} \sqrt[n]{n|x|^n} = |x| \lim_{n\to\infty} \sqrt[n]{n} = |x|\right) < 1$$

and

$$\sum_{n=0}^{\infty} |nx^n| < \infty \implies \sum_{n=0}^{\infty} nx^n < \infty.$$

If $x = -1$, then

$$\lim_{n\to\infty} nx^n = \lim_{n\to\infty} (-1)^n n \neq 0 \implies \sum_{n=0}^{\infty} nx^n = \sum_{n=0}^{\infty} (-1)^n n \not< \infty.$$

If $x = 1$, then

$$\lim_{n\to\infty} nx^n = \lim_{n\to\infty} n \neq 0 \implies \sum_{n=0}^{\infty} nx^n = \sum_{n=0}^{\infty} n = \infty.$$

(c) By the Ratio Test,

$$x \in (-\infty,\infty) \iff \lim_{n\to\infty} \left|\frac{(2n)!(x-10)^{n+1}}{(2n+2)!(x-10)^n}\right| \left(= \lim_{n\to\infty} \frac{|x-10|}{4n^2+5n+2} = 0\right) < 1$$

and

$$\sum_{n=0}^{\infty} \left|\frac{1}{(2n)!}(x-10)^n\right| < \infty \implies \sum_{n=0}^{\infty} \frac{1}{(2n)!}(x-10)^n < \infty.$$

(d) By the Ratio Test,

$$x = 0 \iff \lim_{n\to\infty} \left|\frac{(n+1)!x^{n+1}}{n!x^n}\right| \left(= \lim_{n\to\infty} |(n+1)x| = |x| \lim_{n\to\infty} (n+1) = 0\right) < 1$$

and

$$\sum_{n=0}^{\infty} |n!x^n| < \infty \implies \sum_{n=0}^{\infty} n!x^n < \infty.$$

$\square$

Exercise 3. (Cauchy-Schwartz Inequality) Prove if $\sum |x_n|$ and $\sum |y_n|$ converge, then $\sum x_n y_n$ converges absolutely and

$$\left|\sum_{n=1}^{\infty} x_n y_n\right| \leq \sqrt{\sum_{n=1}^{\infty} |x_n|^2 \sum_{n=1}^{\infty} |y_n|^2}.$$

Proof. Since $\sum |x_n|$ and $\sum |y_n|$ are convergent, $\sum |x_n|^2$ and $\sum |y_n|^2$ are convergent. We want to prove that

$$\sum_{n=1}^{\infty} |x_n y_n| < \infty \implies \sum_{n=1}^{\infty} x_n y_n < \infty.$$

By the Direct Comparison Test,

$$\sum_{n=1}^{\infty} |x_n y_n| = \lim_{m\to\infty} \sum_{n=1}^{m} |x_n y_n| \leq \lim_{m\to\infty} \sum_{n=1}^{m} |x_n| \sum_{n=1}^{m} |y_n| = \sum_{n=1}^{\infty} |x_n| \sum_{n=1}^{\infty} |y_n| < \infty.$$

We want to prove that

$$\left| \sum_{n=1}^{\infty} x_n y_n \right| \leq \sqrt{\sum_{n=1}^{\infty} |x_n|^2 \sum_{n=1}^{\infty} |y_n|^2}.$$

Let

$$X_m = \sum_{n=1}^{m} |x_n|^2 \wedge Y_m = \sum_{n=1}^{m} |y_n|^2 \wedge Z_m = \sum_{n=1}^{m} x_n y_n.$$

Then

$$\sum_{n=1}^{m} |x_n Y_m - y_n Z_m|^2 \geq 0 \implies \sum_{n=1}^{m} (x_n Y_m - y_n Z_m)^2 \geq 0.$$

Without loss of generality, assume $\sum |x_n| = 0$. Then

$$X_m = 0 \wedge Z_m = 0 \implies X_m Y_m \geq |Z_m|^2 \implies \sqrt{X_m Y_m} \geq |Z_m|$$

and

$$\left| \sum_{n=1}^{m} x_n y_n \right| \leq \sqrt{\sum_{n=1}^{m} |x_n|^2 \sum_{n=1}^{m} |y_n|^2}.$$

Without loss of generality, assume $\sum |y_n| > 0$. Then

$$\begin{aligned}
\sum_{n=1}^{m} (x_n Y_m - y_n Z_m)^2 &= \sum_{n=1}^{m} (x_n^2 Y_m^2 - 2 x_n y_n Y_m Z_m + y_n^2 Z_m^2) \\
&= Y_m^2 \sum_{n=1}^{m} |x_n|^2 - 2 Y_m Z_m \sum_{n=1}^{m} x_n y_n + Z_m^2 \sum_{n=1}^{m} |y_n|^2 \\
&= X_m Y_m^2 - Y_m |Z_m|^2 = (X_m Y_m - |Z_m|^2) Y_m
\end{aligned}$$

and

$$\sum_{n=1}^{m} (x_n Y_m - y_n Z_m)^2 \geq 0 \implies (X_m Y_m - |Z_m|^2) Y_m \geq 0;$$

however,

$$Y_m > 0 \implies X_m Y_m - |Z_m|^2 \geq 0 \implies \sqrt{X_m Y_m} \geq |Z_m|^2$$

and

$$\left| \sum_{n=1}^{m} x_n y_n \right| \leq \sqrt{\sum_{n=1}^{m} |x_n|^2 \sum_{n=1}^{m} |y_n|^2}.$$

By the Limit Properties of $\mathbb{R}$,

$$\left|\sum_{n=1}^{\infty} x_n y_n\right| = \lim_{m\to\infty}\left|\sum_{n=1}^{m} x_n y_n\right| \le \lim_{m\to\infty}\sqrt{\sum_{n=1}^{m}|x_n|^2\sum_{n=1}^{m}|y_n|^2}$$
$$= \sqrt{\lim_{m\to\infty}\sum_{n=1}^{m}|x_n|^2\sum_{n=1}^{m}|y_n|^2} = \sqrt{\sum_{n=1}^{\infty}|x_n|^2\sum_{n=1}^{\infty}|y_n|^2}.$$

Thus, $\sum x_n y_n$ converges (absolutely) and $|\sum x_n y_n| \le \sqrt{\sum |x_n|^2 \sum |y_n|^2}$, as needed. $\square$

Exercise 4. Prove that every real number is a cluster point of the set of irrational numbers.

Proof. Let $x \in \mathbb{R}$. We want to prove that x is a cluster point of $\mathbb{R}\setminus\mathbb{Q}$. By the Density of $\mathbb{R}\setminus\mathbb{Q}$, $\forall\epsilon > 0\ \exists i \in (\mathbb{R}\setminus\mathbb{Q})\setminus\{x\}$ such that

$$x - \epsilon < i < x + \epsilon.$$

If $\epsilon > 0$, then

$$(x-\epsilon, x+\epsilon) \cap (\mathbb{R}\setminus\mathbb{Q})\setminus\{x\} \neq \emptyset.$$

Thus, x is a cluster point of $\mathbb{R}\setminus\mathbb{Q}$, as needed. $\square$

Exercise 5. Suppose $S \subset \mathbb{R}$ and c is a cluster point of S. Suppose $f : S \to \mathbb{R}$ is bounded. Show that there exists a sequence $\{x_n\}$ with $x_n \in S\setminus\{c\}$ and $\lim x_n = c$ such that $\{f(x_n)\}$ converges.

Proof. Let $S \subset \mathbb{R}$ and $f : S \to \mathbb{R}$. Suppose c is a cluster point of S and f is a bounded function of $\mathbb{R}$. We want to prove that $\exists$ sequence $x : \mathbb{N} \to S\setminus\{c\}$ with $\lim x_n = c$. Choose

$$\forall n \in \mathbb{N} : x_n \in \left[\left(c - \frac{1}{n}, c + \frac{1}{n}\right) \cap S\right] \setminus \{c\} \neq \emptyset.$$

By the Archimedean Property of $\mathbb{R}$, $\forall\epsilon > 0\ \exists M \in \mathbb{N}\ \forall n \ge M$ such that $M\epsilon > 1$ and

$$|x_n - c| < \frac{1}{n} \le \frac{1}{M} < \epsilon.$$

Thus, $x_n \in S\setminus\{c\}$ and $\lim x_n = c$, as needed. We want to prove that $\exists$ subsequence $y : \mathbb{N} \to S\setminus\{c\}$ such that $\{f(y_n)\}$ is convergent. Given that f is bounded, $\exists N > 0\ \forall x \in S$ such that $|f(x)| \le N$. Then

$$\forall n \in \mathbb{N} : x_n \in S\setminus\{c\} \implies |f(x_n)| \le N.$$

By the Bolzano-Weierstrass Theorem, $\{f(x_n)\}$ has a convergent subsequence; hence, $\exists$ subsequence $\{x_{n_i}\}$ such that $\{f(x_{n_i})\}$ is a convergent sequence, as needed. Given that $\{x_n\}$ is a convergent sequence, $\{x_{n_i}\}$ is a convergent subsequence. Indeed, $\{x_{n_i}\}$ and $\{f(x_{n_i})\}$ are convergent sequences such that $x_{n_i} \in S\setminus\{c\}$ and $\lim x_{n_i} = c$, as needed. $\square$

Exercise 6. Let $S \subset \mathbb{R}$, let c be a cluster point of S, and let $f : S \to \mathbb{R}$.

(a) Assume $\lim_{x\to c} f(x)$ exists. Prove that there exist $\delta > 0$ and $B \ge 0$ such that if $x \in S$ and $0 < |x - c| < \delta$ then $|f(x)| \le B$.

(b) Assume that $\lim_{x\to c} f(x) = \lambda > 0$. Prove that there exists $\delta > 0$ such that if $x \in S$ and $0 < |x - c| < \delta$ then $f(x) > 0$.

Proof. Let $S \subset \mathbb{R}$ and $f : S \to \mathbb{R}$.

(a) Assume c is a cluster point of S such that $\lim_{x \to c} f(x) = \lambda \in \mathbb{R}$. Choose $\epsilon > 0$. Then

$$\exists \delta > 0 \; \forall x \in S : 0 < |x - c| < \delta \implies |f(x) - \lambda| < \epsilon \implies \lambda - \epsilon < f(x) < \lambda + \epsilon.$$

Fix $B = \max\{|\lambda - \epsilon|, |\lambda + \epsilon|\} \geq 0$. Then

$$\exists B \geq 0 \; \forall x \in S \setminus \{c\} : -B \leq \lambda - \epsilon < f(x) < \lambda + \epsilon \leq B \implies |f(x)| \leq B.$$

Indeed,

$$\exists \delta > 0 \wedge B \geq 0 \; \forall x \in S : 0 < |x - c| < \delta \implies |f(x)| \leq B,$$

as needed.

(b) Assume c is a cluster point of S such that $\lim_{x \to c} f(x) = \lambda > 0$. Choose $\epsilon \in (0, \lambda)$. Thus,

$$\exists \delta > 0 \; \forall x \in S : 0 < |x - c| < \delta \implies |f(x) - \lambda| < \epsilon \implies f(x) > \lambda - \epsilon > 0,$$

as required.

$\square$